

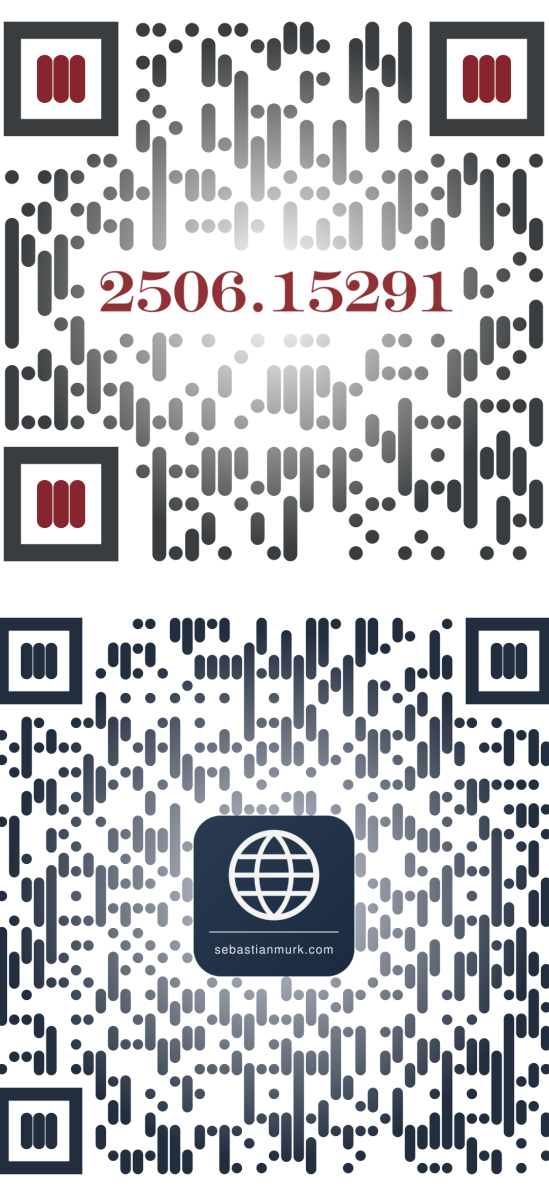
Classical-quantum gravity as quantum gravity in disguise

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Completely positive classical-quantum (CPCQ) dynamics

A classical-quantum (CQ) state is described by a positive operator-valued density $\hat{\rho}(z, t)$ over classical phase space $z = (q, p)$ satisfying

$$\hat{\rho}(z, t) \succeq 0, \quad \int \text{Tr}_Q[\hat{\rho}(z, t)] dz = 1.$$

The classical (C) and quantum (Q) subsystems are recovered by

C — Tracing over the quantum system: Q — Integrating over the classical variables:

$$\varrho(z, t) = \text{Tr}_Q[\hat{\rho}(z, t)] \quad \rho_Q(t) = \int \hat{\rho}(z, t) dz$$

Completely positive (CP) evolution is described by a Kraus transition kernel

$$\hat{\rho}(z, t) = \sum_{\alpha} \int K_{\alpha Q}(z|z', t) \hat{\rho}(z', 0) K_{\alpha Q}^{\dagger}(z|z', t) dz',$$

where each $K_{\alpha Q}(z|z', t)$ combines a classical transition $z' \rightarrow z$ with a quantum operation on Q. The family of Kraus operators satisfies a completeness relation ensuring preservation of total probability.

Quantum backreaction requires decoherence and diffusion

The CP transition amplitudes are characterized by statistical moments $D_{(n)}$ of the phase space increment $\Delta z = z - z'$. Their CQ state averages have the following physical interpretations:

$\langle D_{(0)} \rangle$	quantum decoherence/dissipation
$\langle D_{(1)}^{\text{br}} \rangle_i$	mean backreaction force or phase space drift
$\langle D_{(2)}^{\text{br}} \rangle_{ij}$	backreaction-induced phase space diffusion

Genuine quantum backreaction requires $\langle D_{(0)} \rangle > 0$. Furthermore, complete positivity requires

$$2 \langle D_{(2)}^{\text{br}} \rangle_{ij} \langle D_{(0)} \rangle \geq \langle D_{(1)}^{\text{br}} \rangle_i \langle D_{(1)}^{\text{br}} \rangle_j \quad \text{for every CQ state and all } i, j.$$

➡ **Backreaction, decoherence, and diffusion cannot be chosen independently.**

Main Result: Completely positive classical-quantum dynamics can be embedded into a fully quantum theory

For a discrete or coarse-grained classical label z , its values can be encoded in an orthonormal basis $\{|z\rangle_Z\}_z$ of an auxiliary quantum register Z with projectors $\Pi_z = |z\rangle_Z \langle z|_Z$.

The CQ Kraus operators $K_{\alpha Q}$ then define Kraus operators on QZ,

$$\Xi_{\alpha}(t) = \sum_{z, z'} K_{\alpha Q}(z|z', t) \otimes |z\rangle_Z \langle z'|_Z, \quad \text{and hence the quantum channel } \Gamma_t[X] = \sum_{\alpha} \Xi_{\alpha}(t) X \Xi_{\alpha}^{\dagger}(t), \quad \sum_{\alpha} \Xi_{\alpha}^{\dagger}(t) \Xi_{\alpha}(t) = I_{QZ}.$$

Encoding the initial CQ state as the block-diagonal quantum state

$$\rho_{QZ}(0) = \sum_{z'} \hat{\rho}_Q(z', 0) \otimes \Pi_{z'}, \quad \text{the diagonal blocks of the quantum channel reproduce the CQ evolution exactly: } \langle z| \Gamma_t[\rho_{QZ}(0)] |z\rangle_Z = \hat{\rho}_Q(z, t). \quad \text{The channel output itself need not be Z-diagonal.}$$

The operational restriction to the classical sector is implemented by the Z-pinching map $\mathcal{P}_Z[X] := \sum_z (I_Q \otimes \Pi_z) X (I_Q \otimes \Pi_z)$, $\rho_{QZ}(t) := \mathcal{P}_Z[\Gamma_t[\rho_{QZ}(0)]] = \sum_z \hat{\rho}_Q(z, t) \otimes \Pi_z$.

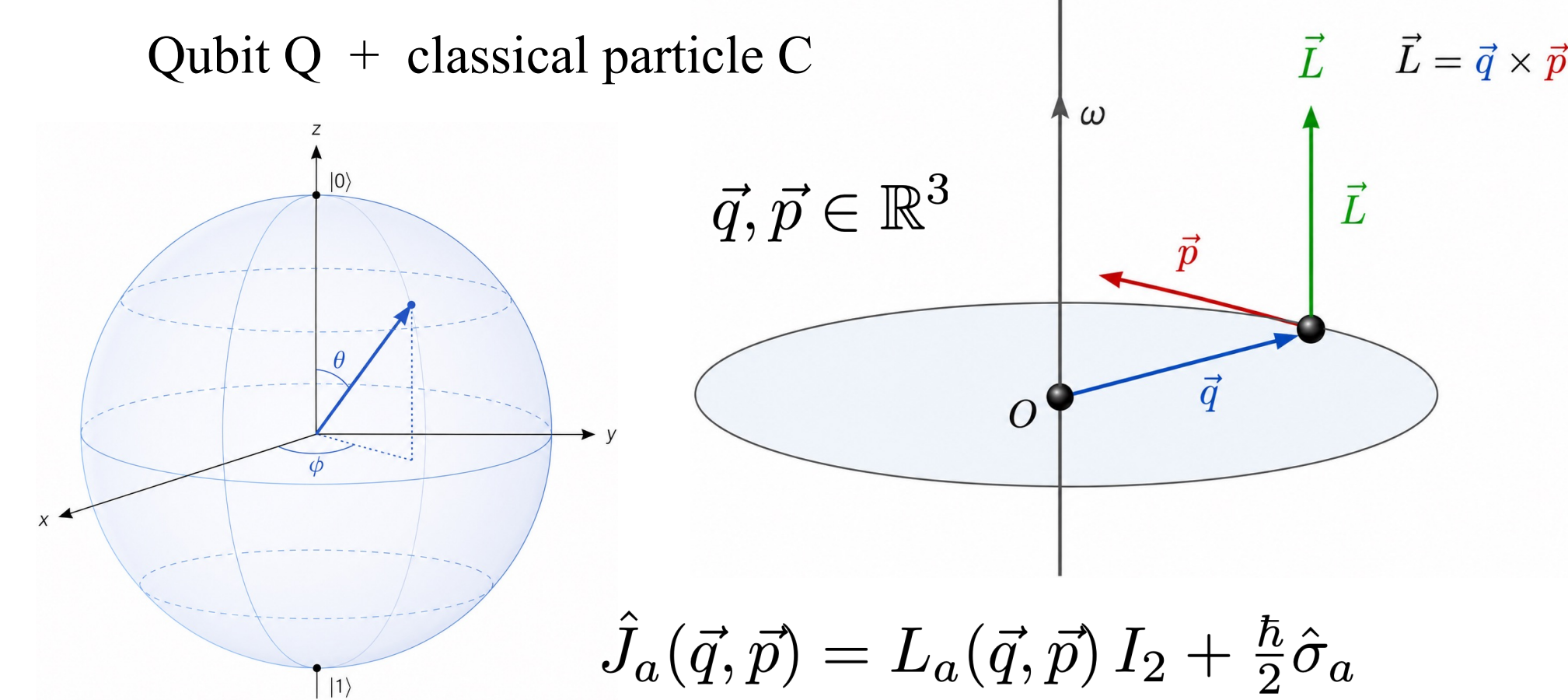
Every CPTP map Γ_t admits a unitary Stinespring dilation on QZE. The construction can therefore be summarized as

unitary dynamics on QZE $\xrightarrow{\text{Tr}_E}$ CPTP map Γ_t on QZ $\xrightarrow[\mathcal{A}_Z \text{ on observables}]{\mathcal{P}_Z \text{ on states}}$ CPCQ dynamics

Thus, CPCQ dynamics is recovered exactly as the operationally restricted Z-diagonal sector of a CPTP map on QZ. ➡ CPCQ gravity need not be irreducibly hybrid.

Second Result: Rotational symmetry without angular momentum conservation

Closed CPCQ toy model



Rotationally invariant CP relaxation

With decoherence rate $\kappa > 0$, the model drives every initial CQ state exponentially toward the stationary state $\hat{\rho}_{\infty}$:

$$\hat{\rho}(t) = e^{-\kappa t} \hat{\rho}(0) + (1 - e^{-\kappa t}) \hat{\rho}_{\infty},$$

$$\hat{\rho}_{\infty}(\vec{q}, \vec{p}) = \frac{I_2}{2} \rho_f(|\vec{q}|, |\vec{p}|, \vec{q} \cdot \vec{p}).$$

Because ρ_f depends only on rotational scalars, the equation of motion is rotationally invariant. Choosing $\rho_f(\vec{q}, \vec{p}) = \delta^3(\vec{q}) \delta^3(\vec{p})$ gives a stationary state with $\vec{L}_{\infty} = 0$.

Symmetry does not imply conservation

For CP dynamics, an observable is conserved for all initial states precisely when it lies in the kernel of the adjoint Liouvillian:

$$\frac{dA}{dt} = 0 \iff \mathcal{L}^{\dagger}(A) = 0$$

$$\langle J_a(t) \rangle = e^{-\kappa t} \langle J_a(0) \rangle$$

Hence $\langle J_a(t) \rangle \rightarrow 0$ as $t \rightarrow \infty$, despite rotational invariance of the equation of motion.

Scope and interpretation of the embedding theorem

The embedding theorem is an existence result:

CPCQ dynamics can be realized as an operationally restricted sector of a fully quantum theory.

Operational classicality

Allowed observables on Z belong to the commutative pointer algebra

$$\mathcal{A}_Z := \text{span}\{\Pi_z\}_z, \quad \Pi_z = |z\rangle_Z \langle z|_Z.$$

The Z-pinching map $\mathcal{P}_Z$ removes off-diagonal Z blocks without changing expectation values of the allowed observables. Thus, z is operationally classical, even though QZ is fully quantum.

Keep in mind: **Quantum completion $\neq$ canonical quantization**

Noncanonical quantum completion

For gravity $z = (h, \pi)$, and both phase space variables are represented diagonally in the same pointer basis:

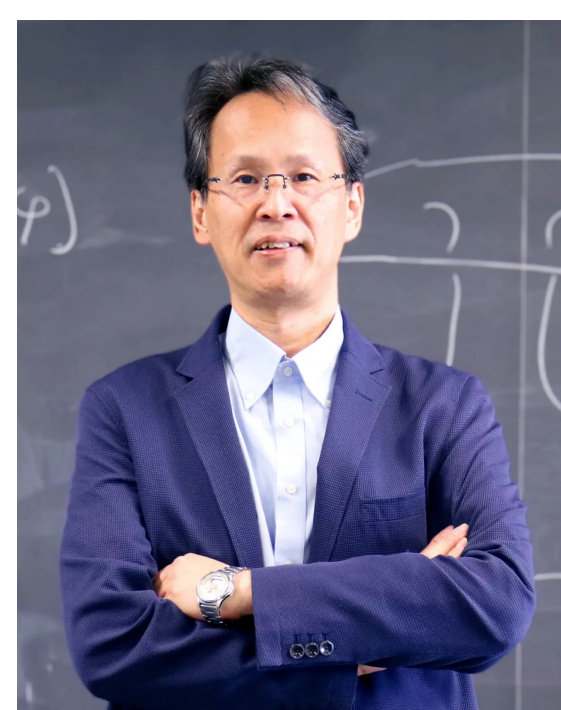
$$h_Z = \sum_z h(z) \Pi_z, \quad \pi_Z = \sum_z \pi(z) \Pi_z, \quad [h_Z, \pi_Z] = 0.$$

However, the construction does not select a unique microscopic QZE completion, and its physical consistency (e.g., compatibility with GR/QFT and conservation laws) remains to be investigated.

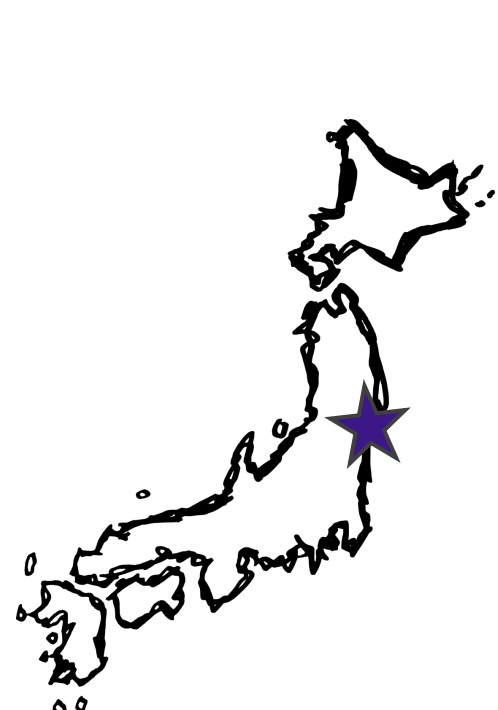
Theory	Symmetry of EOM	Local Conservation	Global Conservation	Noether's Theorem
Classical Mechanics	✓	n/a	✓	✓
Classical Field Theory	✓	✓	✓	✓
Quantum Mechanics (Closed Systems)	✓	n/a	✓	✓
Open Quantum Systems (Lindblad)	✓	⚠ (dissipation, decoherence)	✗ (only full system + bath)	✗
General Relativity	✓	✓	⚠ (not well-defined)	✗
Quantum Field Theory (Flat Spacetime)	✓	✓	✓ (if no quantum anomaly)	✓ (if no quantum anomaly)
Quantum Field Theory (Curved Spacetime)	⚠ (depends on background)	⚠ (anomalies, backreaction)	✗ (no global Killing vector)	⚠ (trace anomaly)
Classical-Quantum Hybrid Models	✓	✗ (no continuity equation)	✗	✗ (no global action)

TABLE I: Symmetries, conservation laws, and Noether's theorem across different theoretical frameworks.

Teamwork makes the dream work: Collaborators and affiliations



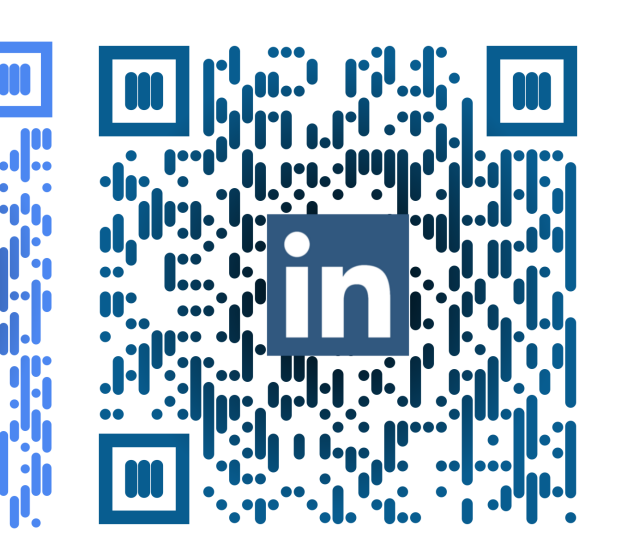
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